

Langlin Huang

Email: h.langlin@wustl.edu | LinkedIn: langlin-huang | Tel: 206-643-8598 | GitHub: shrango.github.io

EDUCATION

- **Washington University in St. Louis (WashU)** *St. Louis, U.S.*
Ph.D. Student in Computer Science - GPA: 4.0/4.0 Aug. 2024 - Jun. 2029 (expected)
- **Institute of Computing Technology, Chinese Academy of Sciences (ICT/CAS)** *Beijing, China*
M.E. in Computer Technology - GPA: 3.75/4.0, Outstanding Dissertation Sep. 2021 - Jun. 2024
- **University of International Business and Economics (UIBE)** *Beijing, China*
B.E. in Data Science and Big Data Technology - GPA: 3.74/4.0 (Rank:2/147) Sep. 2017 - Jun. 2021

RESEARCH INTERESTS

- Large Language Model Reasoning & Post-Training
- Large Language Model Efficiency

PUBLICATIONS

- **PosS: Position Specialist Generates Better Draft for Speculative Decoding**
Langlin Huang, Chengsong Huang, Jixuan Leng, Di Huang, Jiaxin Huang
ICLR 2026 on submission [Paper] [Code]
- **Efficient test-time scaling via self-calibration**
Chengsong Huang, Langlin Huang, Jixuan Leng, Jiacheng Liu, Jiaxin Huang
ICLR 2026 on submission [Paper][Code]
- **Divide, Reweight, and Conquer: A Logit Arithmetic Approach for In-Context Learning**
Chengsong Huang, Langlin Huang, Jiaxin Huang
EACL 2026 oral [Paper][Code]
- **CrossWordBench: Evaluating the Reasoning Capabilities of LLMs and LVLMS with Controllable Puzzle Generation**
Jixuan Leng, Chengsong Huang, Langlin Huang, Bill Yuchen Lin, William W Cohen, Haohan Wang, Jiaxin Huang
COLM 2025 [Paper][Code] [Dataset]
- **MoCE: Adaptive Mixture of Contextualization Experts for Byte-based Neural Machine Translation**
Langlin Huang, Mengyu Bu, Yang Feng
NAACL 2025 oral [Paper] [Code]
- **BayLing: Bridging Cross-lingual Alignment and Instruction Following through Interactive Translation for Large Language Models**
Shaolei Zhang, Qingkai Fang, Zhuocheng Zhang, Zhengrui Ma, Yan Zhou, Langlin Huang, Mengyu Bu, Shangdong Gui, Yunji Chen, Xilin Chen, Yang Feng
Technical Report [Paper] [Code]
- **Integrating Multi-scale Contextualized Information for Byte-based Neural Machine Translation**
Langlin Huang, Yang Feng
ACL findings, 2024 [Paper][Code]
- **Enhancing Neural Machine Translation with Semantic Units**
Langlin Huang, Shuhao Gu, Zhuocheng Zhang, Yang Feng
EMNLP findings, 2023 [Paper][Code]
- **Automatic Construction of a Depression-Domain Lexicon Based on Microblogs: Text Mining Study**
Genghao Li, Bing Li, Langlin Huang, Sibing Hou
JMIR medical informatics, 2020, Vol 8. Jun. 2020 [Paper]

ACADEMIC RESEARCH EXPERIENCE

- **Accelerating LLM Inference Speed via Improved Speculative Decoding** *Feb. 2025 - Jul. 2025*
Proposed a simple and effective method to significantly improve over the strong existing speculative decoding approach. (ICLR 2026 on submission)
 - Resolving the inter-position misalignment in EAGLE via disentangling the single draft model. This improves prediction accuracy at all draft positions and extends the average acceptance length by 0.5 on average.
 - Speeding up LLM inference by about **3.3x**, **10%** faster than EAGLE-3, the previous SOTA approach.
 - Effortless deployment on vLLM. With one A100 GPU, the throughput reaches **1062 tokens/seconds** on code generation benchmark, accelerating vLLM inference by **3.57x**.

•Improving LLM Reasoning via Efficient Test-time Scaling

Dec. 2024 - Feb. 2025

Proposed to incorporate confidence into test-time scaling to improve LLM reasoning, significantly improving prediction accuracy and saving samples used. (ICLR 2026 on submission)

- Applying LLM’s confidence as the weight for answer aggregation in test-time scaling, improving accuracy by **2.4** over standard self-consistency approach.
- Leveraging self-consistency with confidence for LLM calibration. This corrects the severe over-confidence problem in LLMs, reducing expected calibration error (ECE) by up to **80%**.
- Improving test-time scaling efficiency with the calibrated confidence, **saving 94.2%** samples to reach the same accuracy as standard self-consistency.

INDUSTRY EXPERIENCE

•Search Relevance Team, ByteDance

Jan. 2024 - Apr. 2024

Algorithm Engineer

Proposed an adapter module to bridge the relevance model used in production with an LLM. The adapter transformed representations from the traditional model into LLM-compatible hidden states, enabling:

- Joint utilization of pretrained traditional models and powerful LLM reasoning capabilities.
- Robust handling of out-of-vocabulary and multilingual inputs.
- A representation model plus LLM framework extensible to user-isolated data or multimodal input scenarios.

LANGUAGES PROFICIENCY

Master:Python, Shell, LaTeX, C++

Proficient:JAVA, R, C

COMMUNITY SERVICES

Top-reviewer: NeurIPS 2025

Reviewer: NeurIPS, ICLR, COLM, ACL, ACL Rolling Review, TALLIP